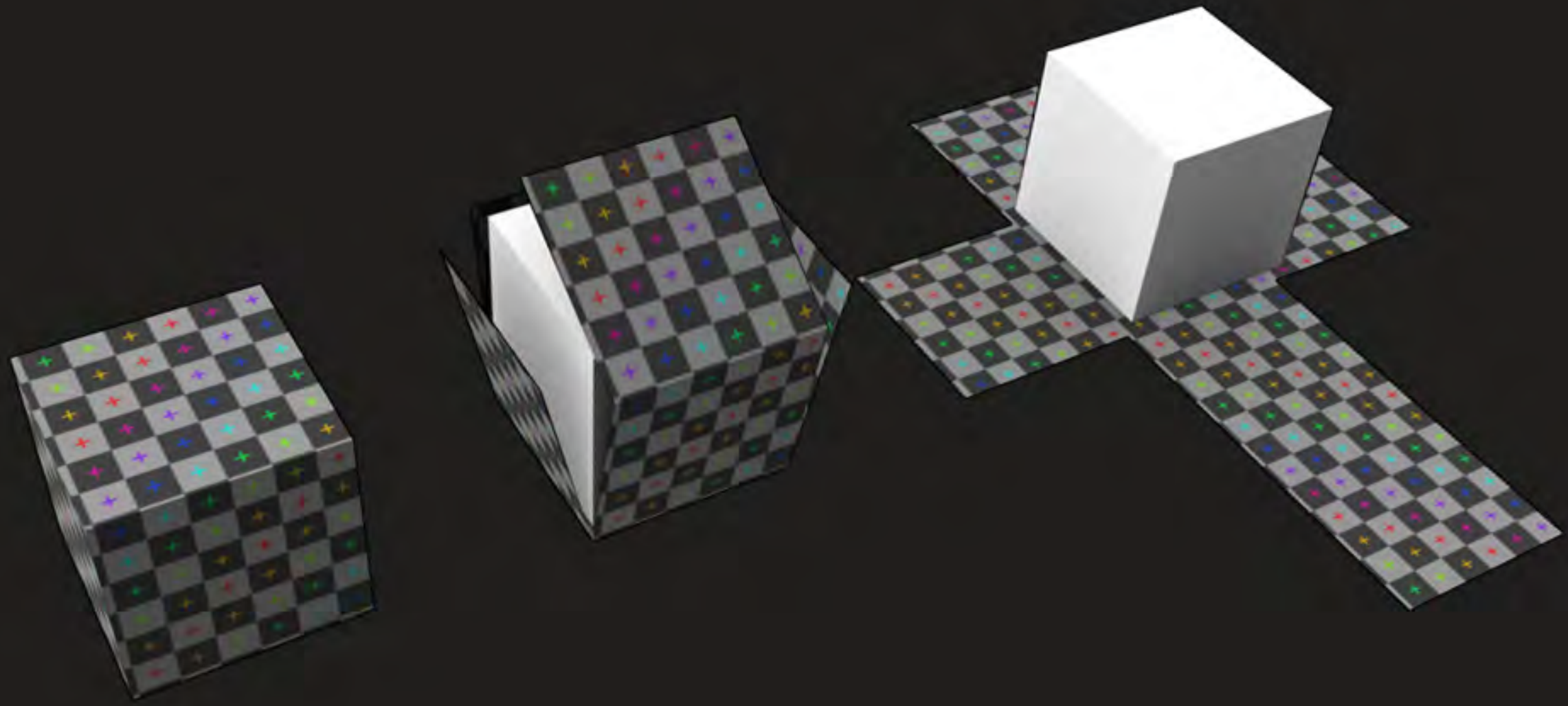
CS 412 / 512
Introduction to Computer Graphics

Zhu Wang
Assistant Professor of Computer Science



COMPUTER SCIENCE

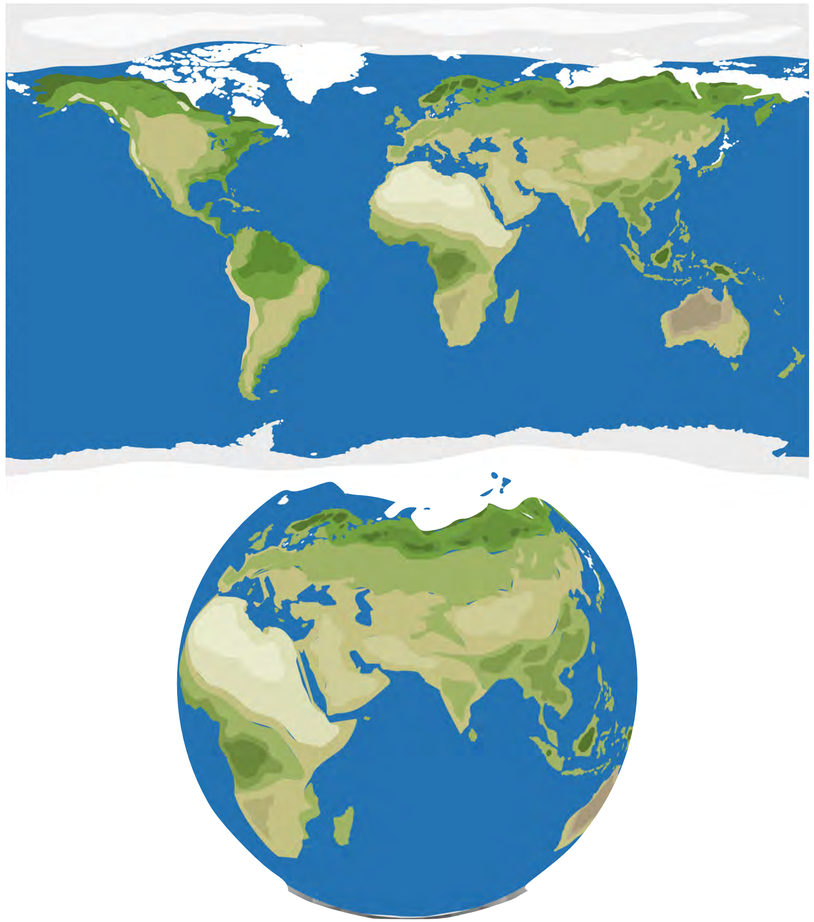
Texture & Texture Mapping

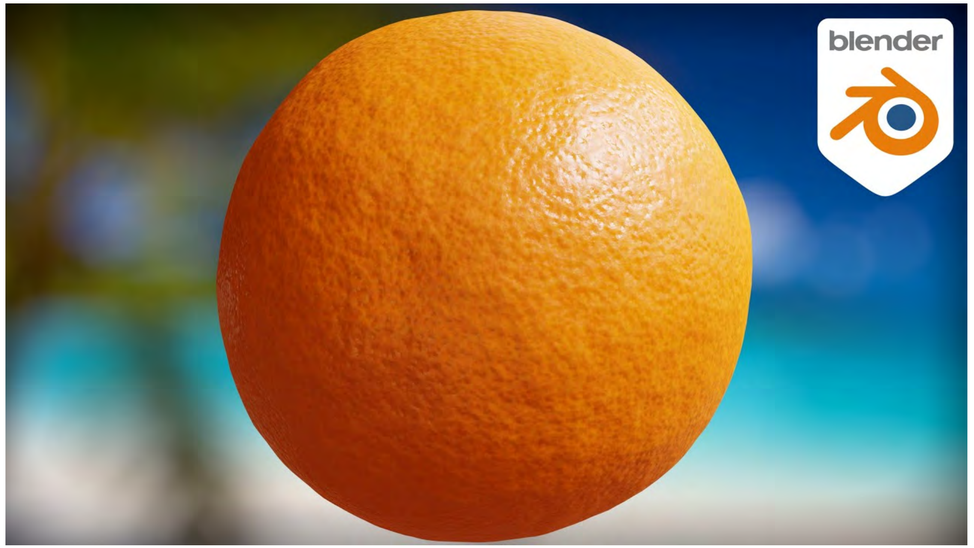


[source: Wikipedia]

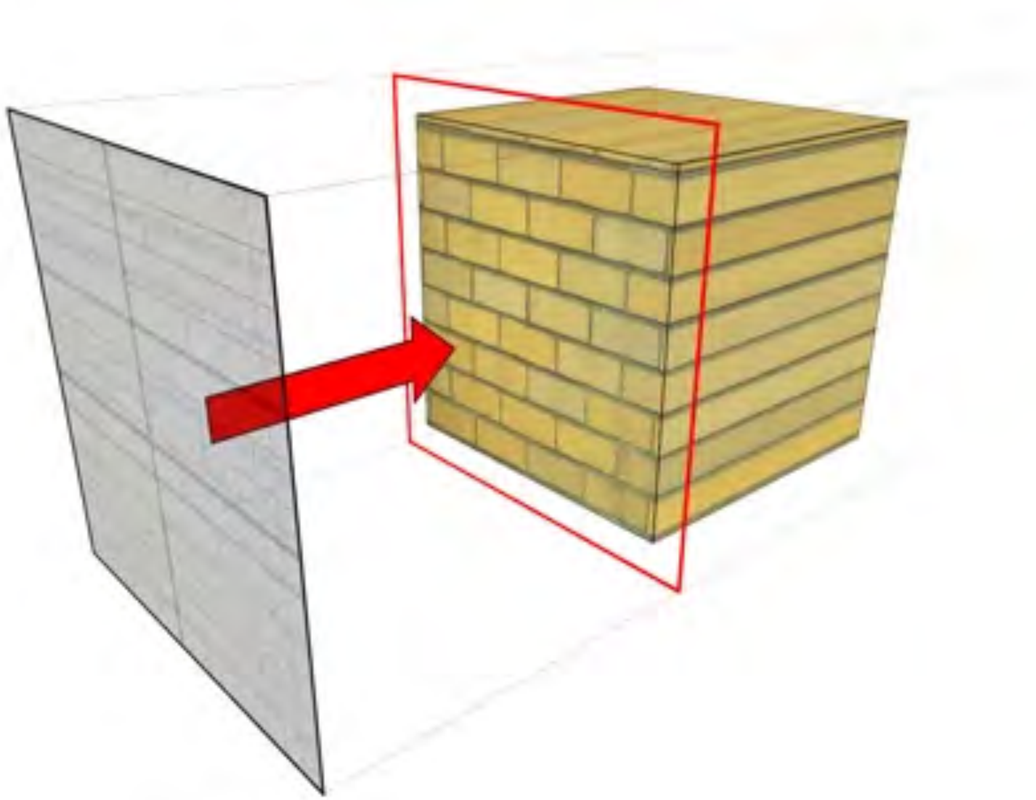
Texture

Surface information stored as an image (pixels) that is applied to models

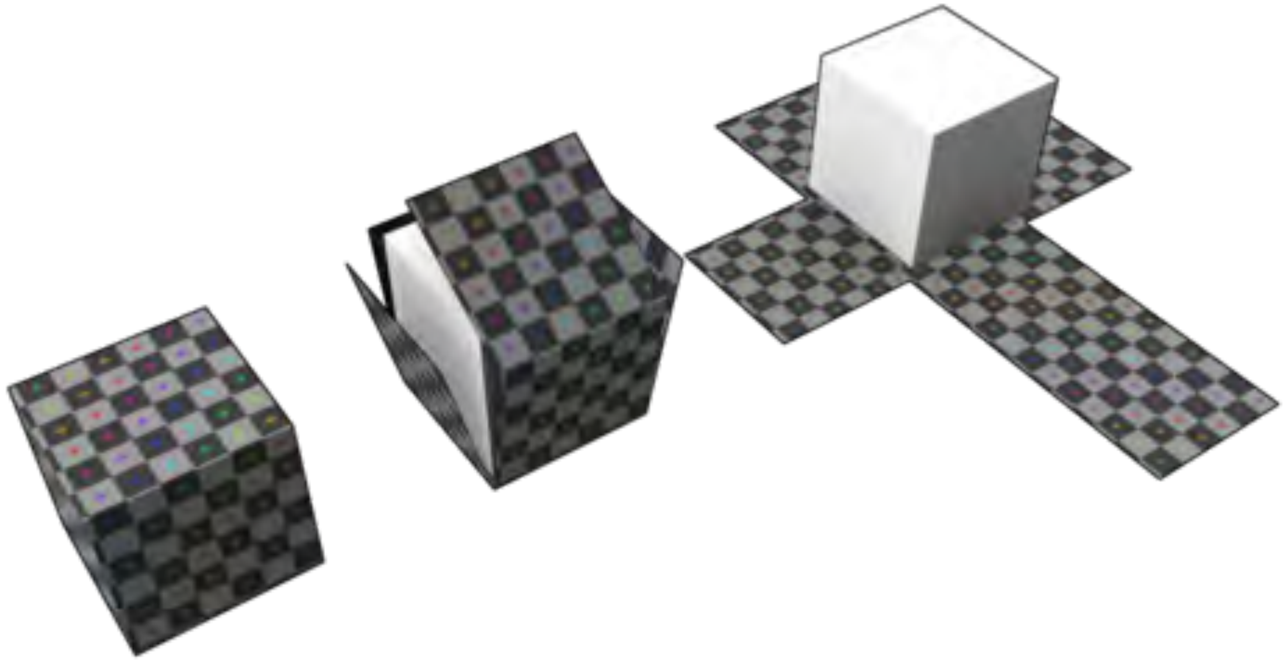




Texture Mapping

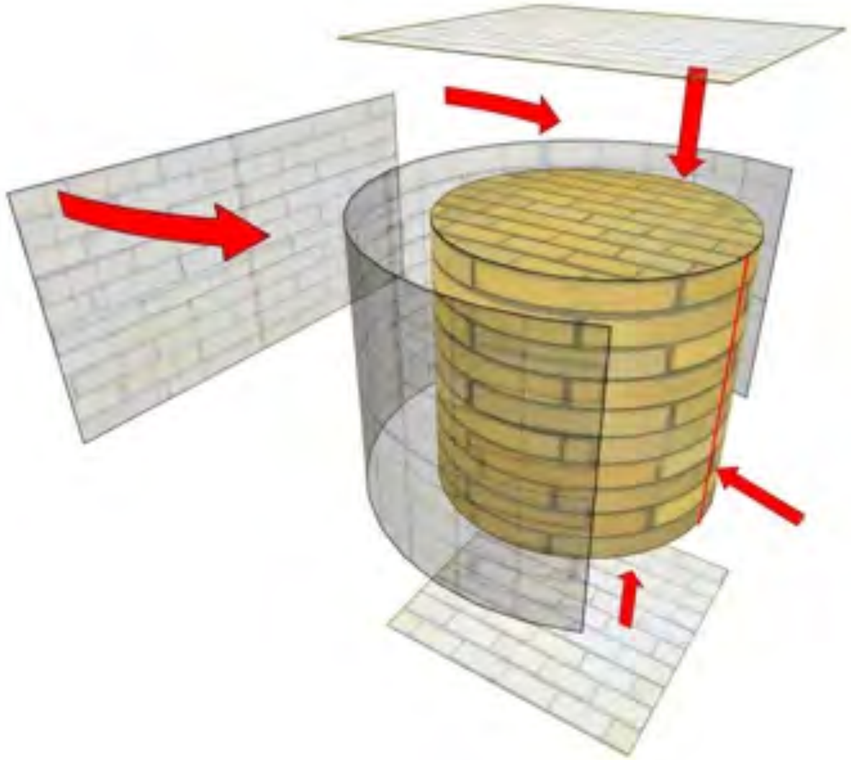


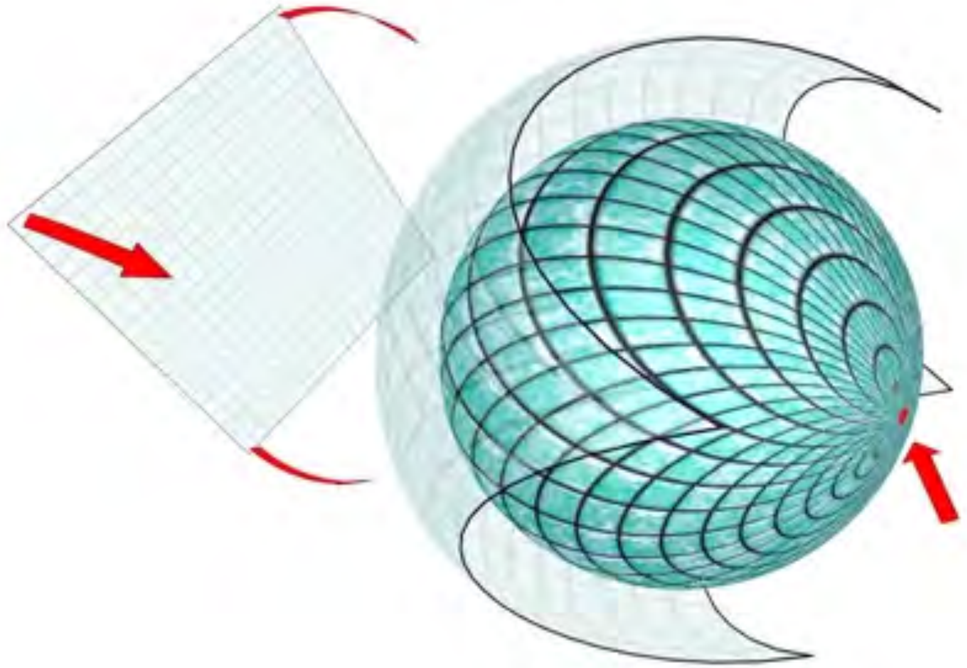
[source: Disguise.one]



[source: Wikipedia]

Texture Mapping





[source: Disguise.one]

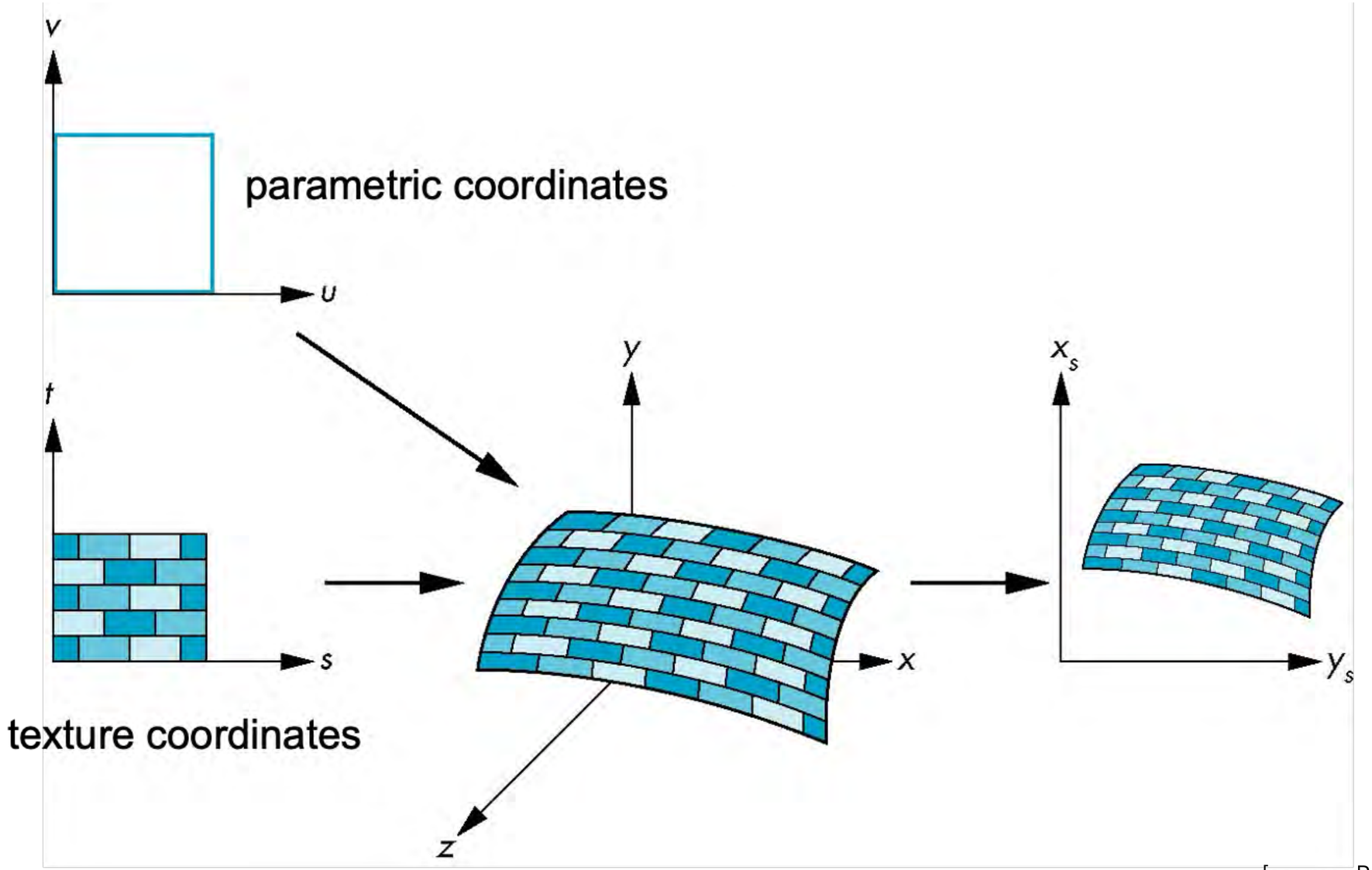
Texture Mapping





[source: Disguise.one]

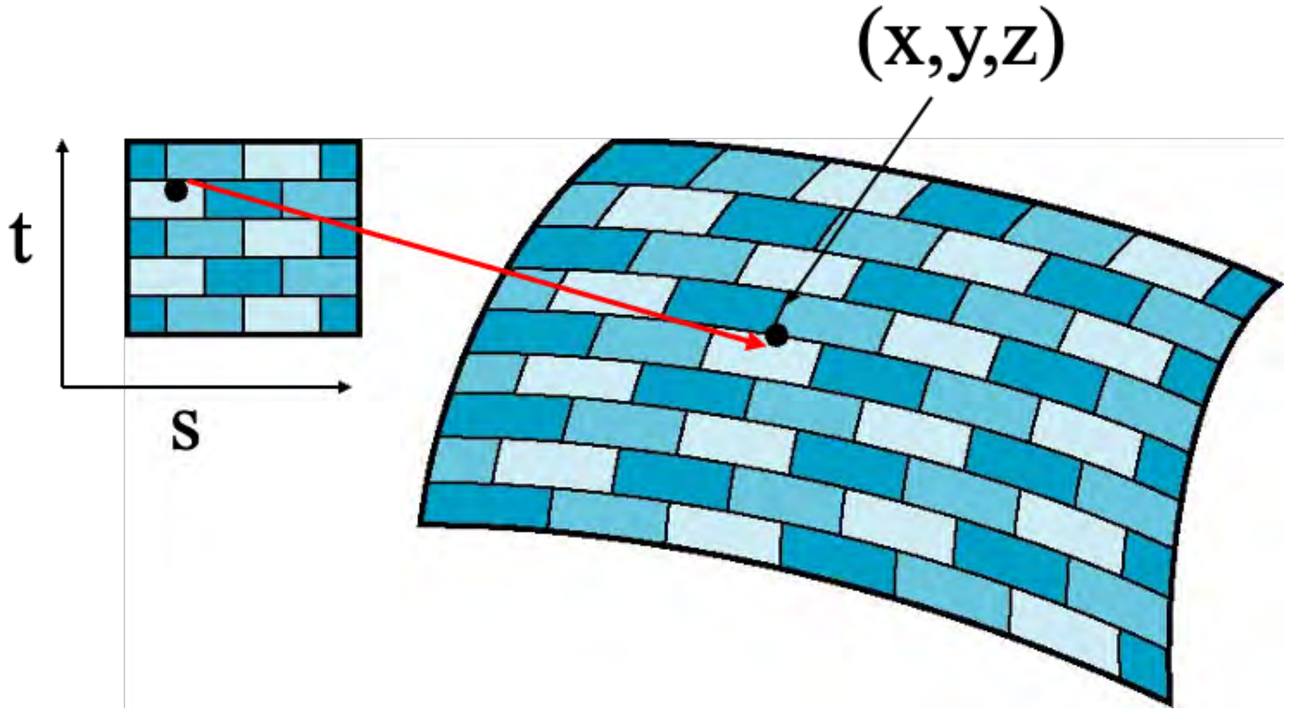
Texture Mapping



[source: Prof. Edward Angel]

Mapping Functions

Forward and backward mapping



[source: Prof. Edward Angel]

Forward Mapping

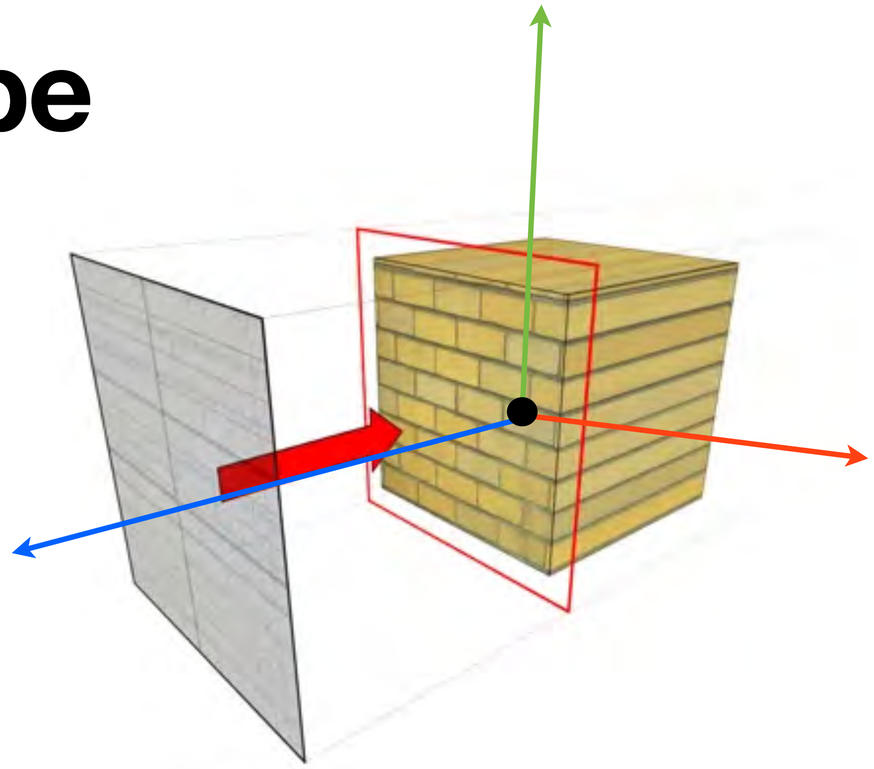
uv → Pixel

Backward Mapping

Pixel → World/Object Space point → uv

Texture Mapping on Cube





We need $u, v \in [0,1]$

$u = 0.5+$ Scaled surface coordinate

$v = 0.5+$ Scaled surface coordinate

For +Z face of a unit cube centered at the origin, $u = x + 0.5$ $v = y + 0.5$

Texture Mapping on Sphere



$$u = \frac{\theta}{2\pi} \qquad v = \frac{\phi}{\pi}$$

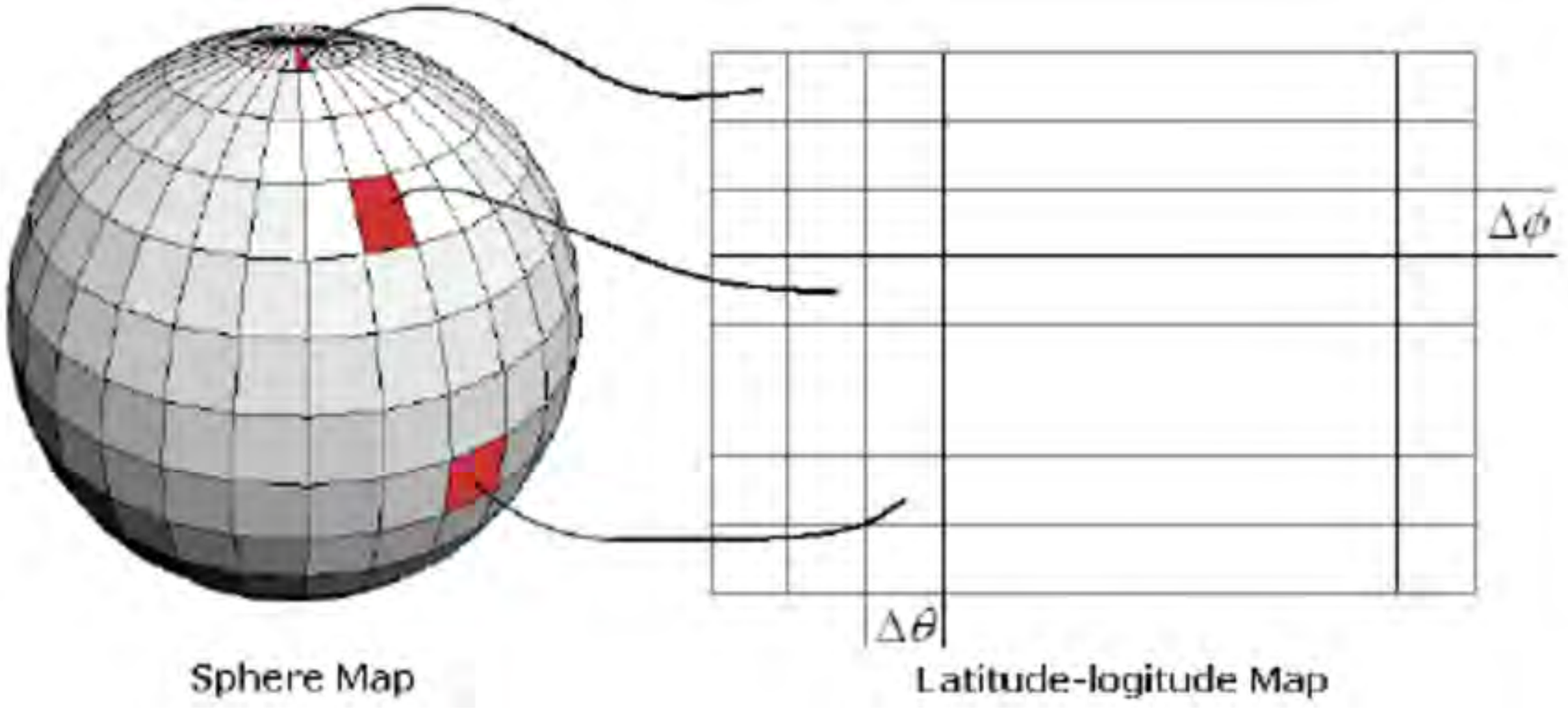
$$u = \frac{arctan2(y, x)}{2\pi} + 0.5 \qquad v = \frac{arcsin(z)}{\pi} + 0.5$$

Shifting u by 0.5 when $\theta \in [-\pi, \pi]$

```
float u = atan(p.y, p.x) / (2.0 * PI) + 0.5;

float v = asin(p.z)/PI + 0.5;
```

Texture Mapping on Sphere



[source: Qi Duan]

A simpler mapping when the mesh is a perfect latitude/longitude grid:

texcoords.push(u_step / uSegments, 1 - (v_step / vSegments));

Texture Mapping on Cylinder



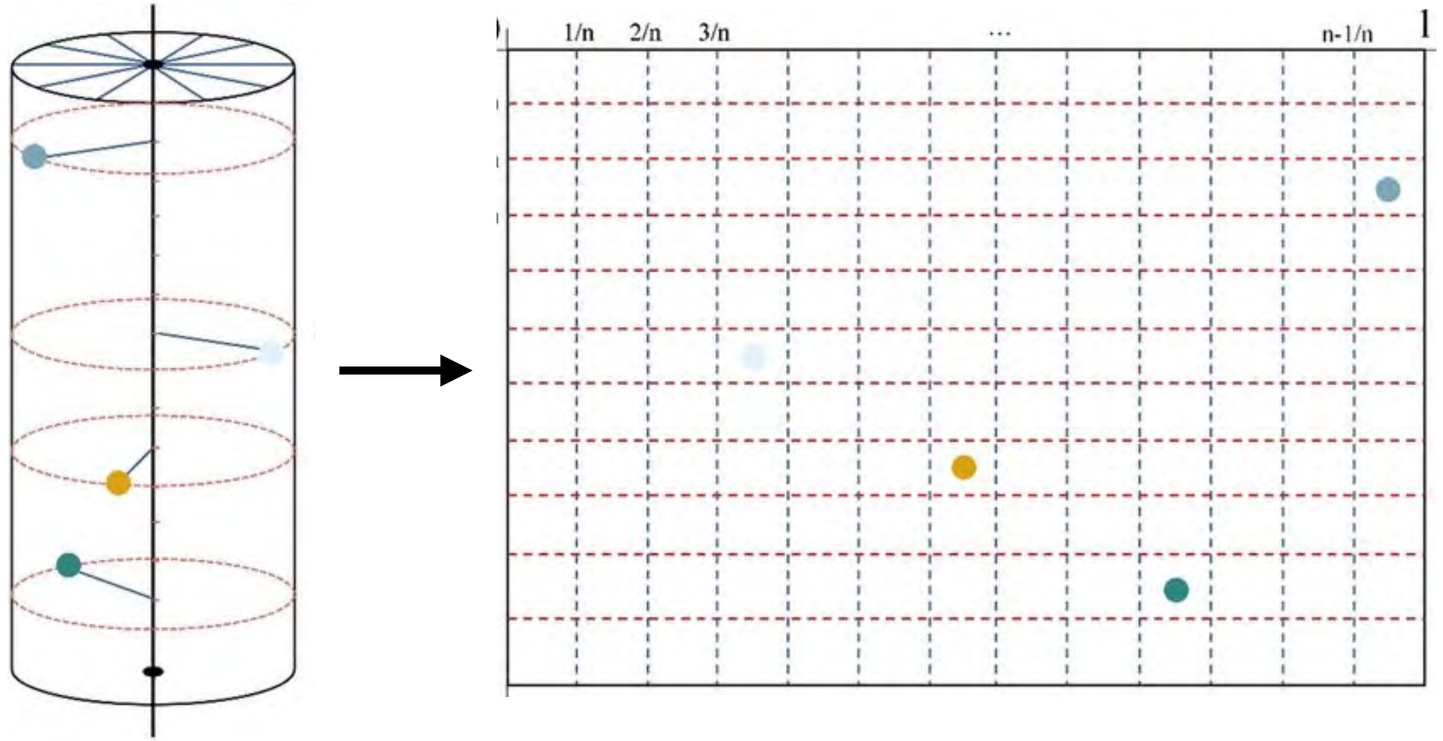
$$u = \frac{\theta}{2\pi} \qquad v = \frac{y - y_{min}}{y_{max} - y_{min}}$$

Shifting u by 0.5 when $\theta \in [-\pi, \pi]$

```
float u = atan(pos.y, pos.x) / (2.0 * PI) + 0.5;

float v = (pos.z - y_min) / (y_max - y_min);
```

Texture Mapping on Cylinder



[source: Zhang Chen et al.]

simpler mapping

texcoords.push(u_step / uSegments, y/height);

Texture Mapping on Cone



$$u = \frac{\theta}{2\pi} \qquad v = \frac{y - y_{min}}{y_{max} - y_{min}}$$

Shifting by 0.5 to make $u, v \in [0,1]$

```
float u = atan(pos.y, pos.x) / (2.0 * PI) + 0.5;

float v = (pos.z - y_min) / (y_max - y_min);
```

simpler mapping

```
texcoords.push(u_step / uSegments, y/height);
```

Texture Mapping on Torus



$$u = \frac{\theta}{2\pi} \qquad v = \frac{\phi}{2\pi}$$

Shifting by 0.5 to make $u, v \in [0,1]$

float u = atan(pos.y, pos.x) / (2.0 * PI) + 0.5;

float v = atan(z, sqrt(x*x+y*y) - R) / (2.0 * PI) + 0.5;

simpler mapping

texcoords.push(u_step / uSegments, v_step/vSegments);

Texture Mapping Pipeline

Create and bind a texture object

```
const texture = gl.createTexture();
gl.bindTexture(gl.TEXTURE_2D, texture);
```

Provide texture information

```
gl.texImage2D(gl.TEXTURE_2D, 0, gl.RGBA, gl.RGBA,
gl.UNSIGNED_BYTE,imageElement);
```

Wrapping mode (tiling)

```
gl.texParameteri(gl.TEXTURE_2D, gl.TEXTURE_WRAP_S, gl.REPEAT);
gl.texParameteri(gl.TEXTURE_2D, gl.TEXTURE_WRAP_T, gl.REPEAT);
```

Texture Mapping Pipeline

Activate texture unit and bind a texture

```
gl.activeTexture(gl.TEXTURE0 + unit); //unit is 0, 1, 2 …
gl.bindTexture(gl.TEXTURE_2D, texture);
gl.uniform1i(gl.getUniformLocation(program, "uTex"), unit);
```

Textures in shaders

```
in  vec2 aTexCoord;
out vec2 vTexCoord;

void main() {
    vTexCoord = aTexCoord;
        …
}
```

```
uniform sampler2D uTexA;// texture A
uniform sampler2D uTexB;// texture B
out vec4 outColor;
void main() {
    vec4 colorA = texture(uTexA, vTexCoord);
    vec4 colorB = texture(uTexB, vTexCoord);
    //blend textures
    outColor = mix(colorA, colorB, 0.5);
}
```

Texture Mapping Pipeline

Combine textures with lighting and shading

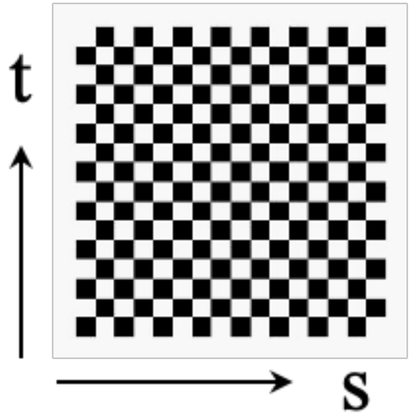
```
vec4 texColor = texture(uSampler, vTexCoord);
vec3 finalColor = (ambient + diffuse + specular) * texColor.rgb;
gl_FragColor = vec4(finalColor, texColor.a);
```

Or

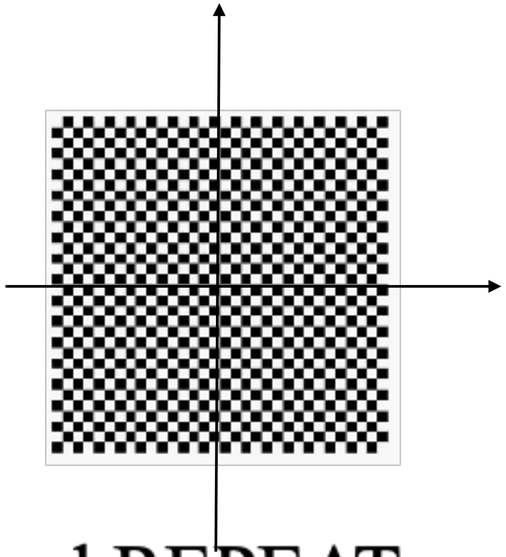
```
vec4 texColor = texture(uSampler, vTexCoord);
vec3 finalColor = (ambient + diffuse) * texColor.rgb + specular;
gl_FragColor = vec4(finalColor, texColor.a);
```

Texture Mapping

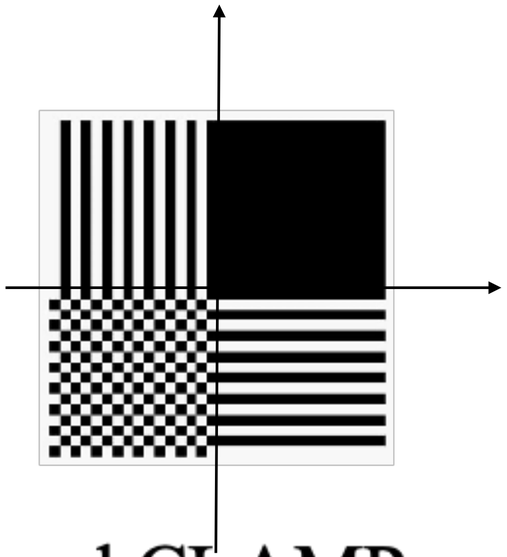
Wrapping mode



texture



gl.REPEAT
wrapping



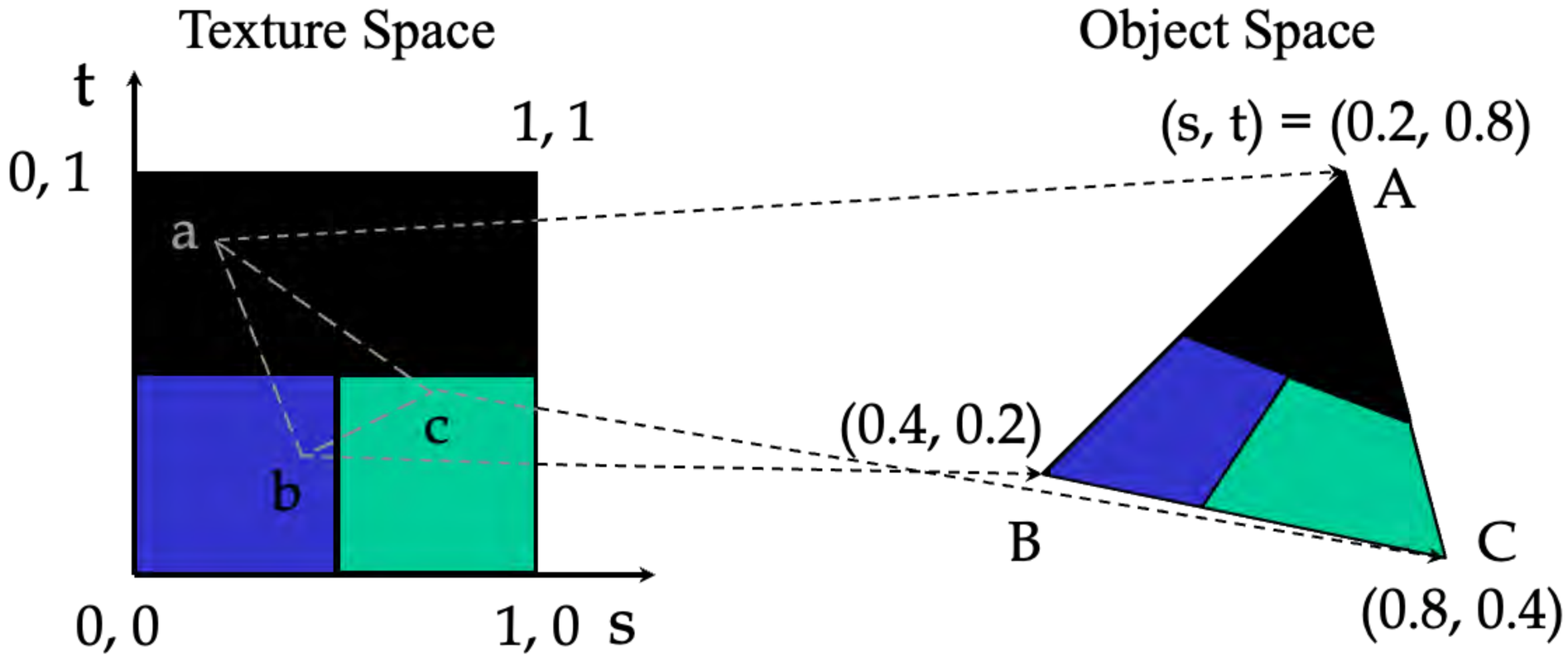
gl.CLAMP
wrapping

gl.REPEAT - tiling

gl.CLAMP - Clamping:
if s,t > 1 use 1, if s,t <0 use 0

[source: Prof. Edward Angel]

Texture Mapping



[source: Prof. Edward Angel]

Texture Mapping

Mapping animations with GIFs or videos

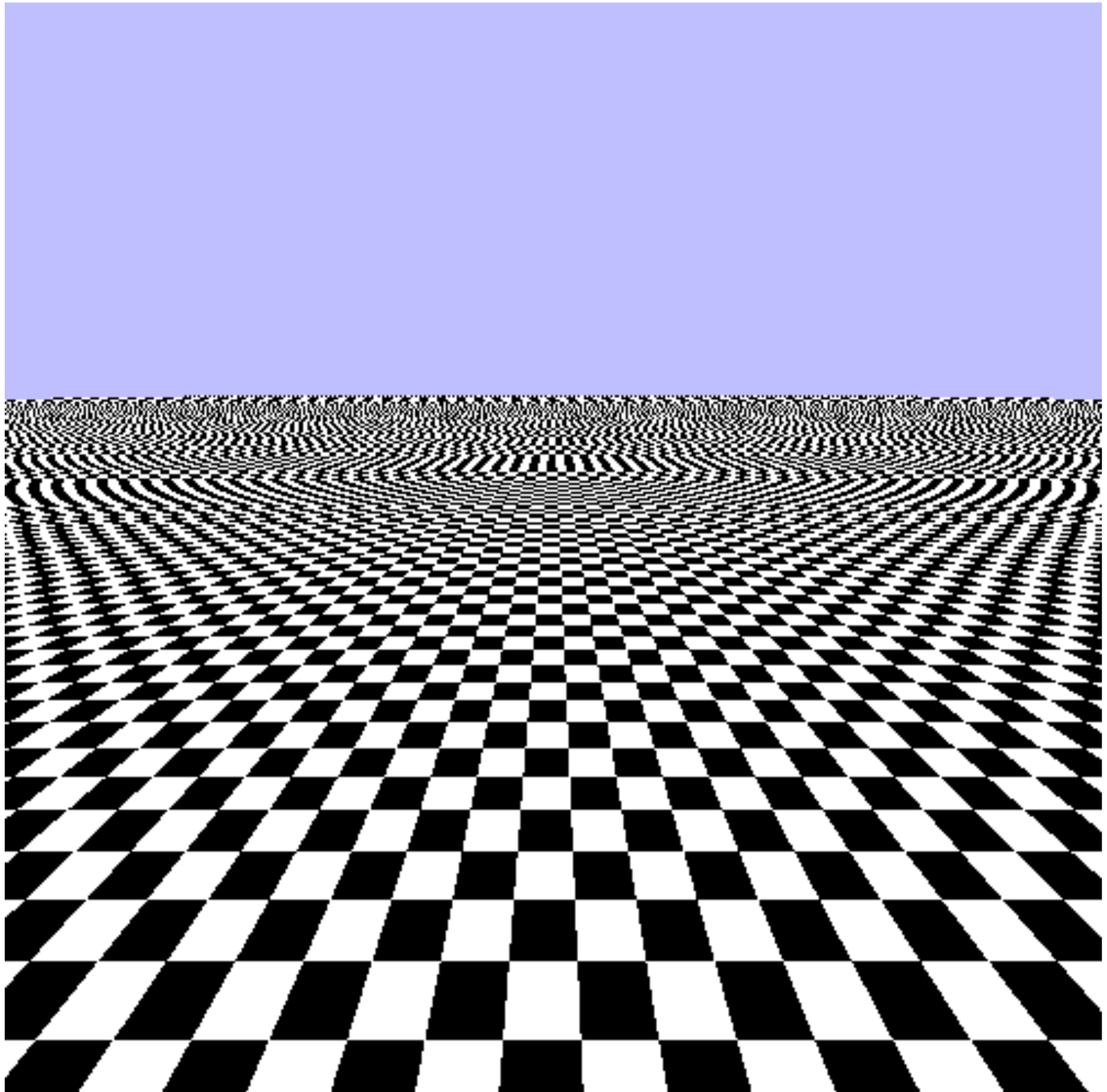



```
videoElement = document.createElement('video');
videoElement.src = url;
videoElement.loop = true;
videoElement.muted = true;
videoElement.play();
textureSource = videoElement;
```

```
if (videoElement && videoElement.readyState >=
                videoElement.HAVE_CURRENT_DATA) {
    gl.bindTexture(gl.TEXTURE_2D, texture);
    gl.texImage2D(gl.TEXTURE_2D, 0, gl.RGBA, gl.RGBA,
                gl.UNSIGNED_BYTE, videoElement);
                …}
```

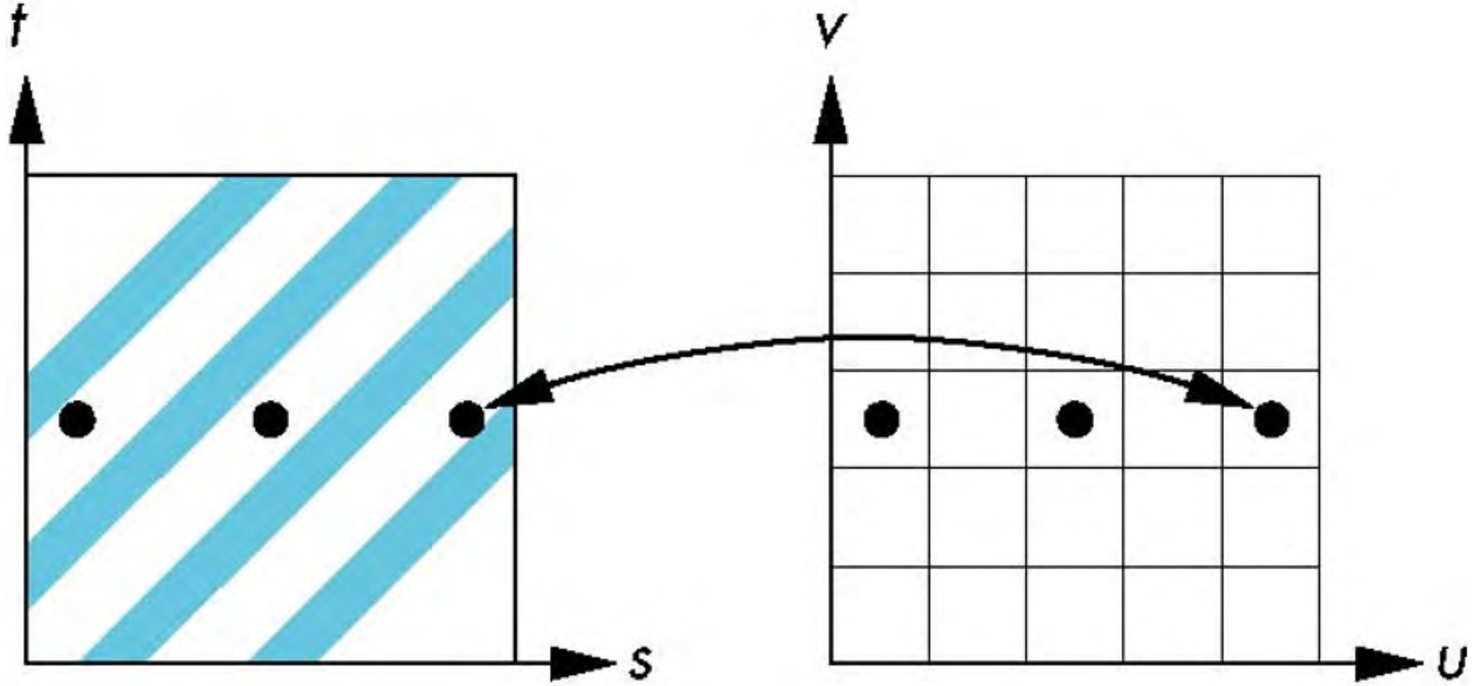
Aliasing, Filtering & Mipmap

Aliasing



[source: Sophismata]

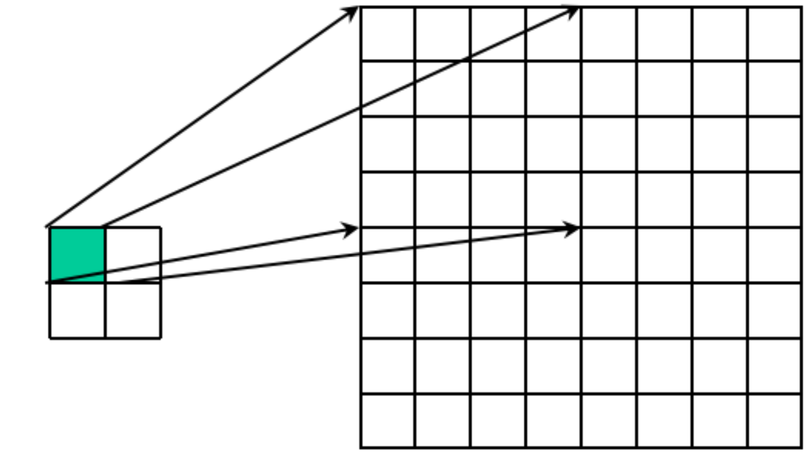
Aliasing



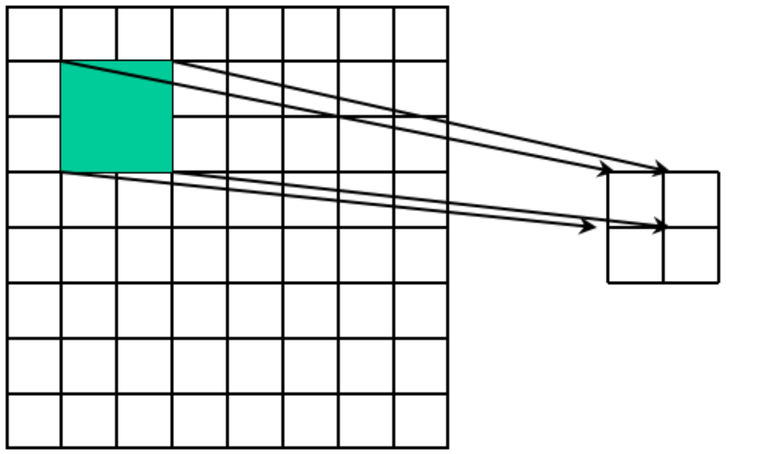
Point sampling of the texture can lead to aliasing errors

[source: Prof. Edward Angel]

Magnification and Minification



Magnification



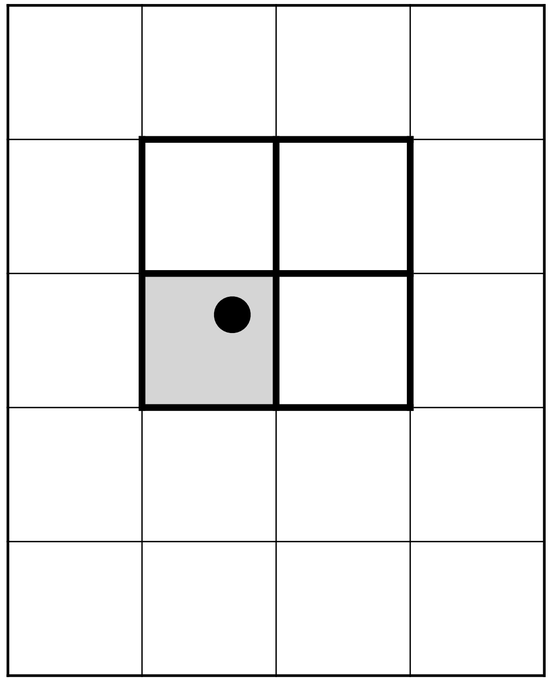
Minification

Magnification: more than one pixel can cover a texel

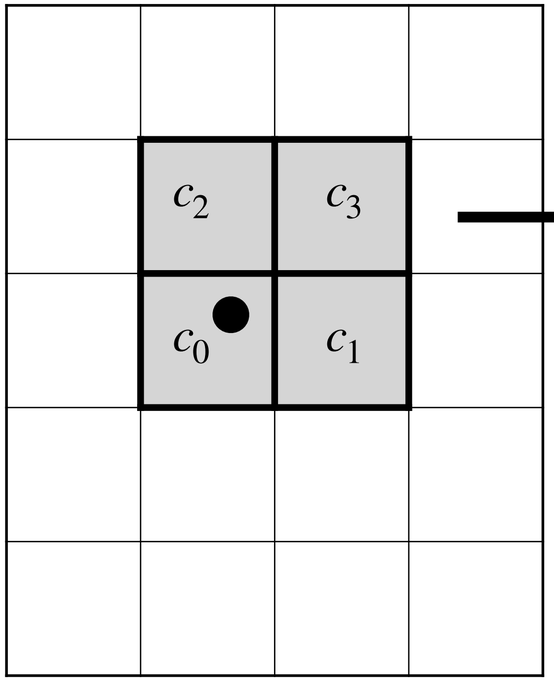
Minification: more than one texel can cover a pixel

[source: Prof. Edward Angel]

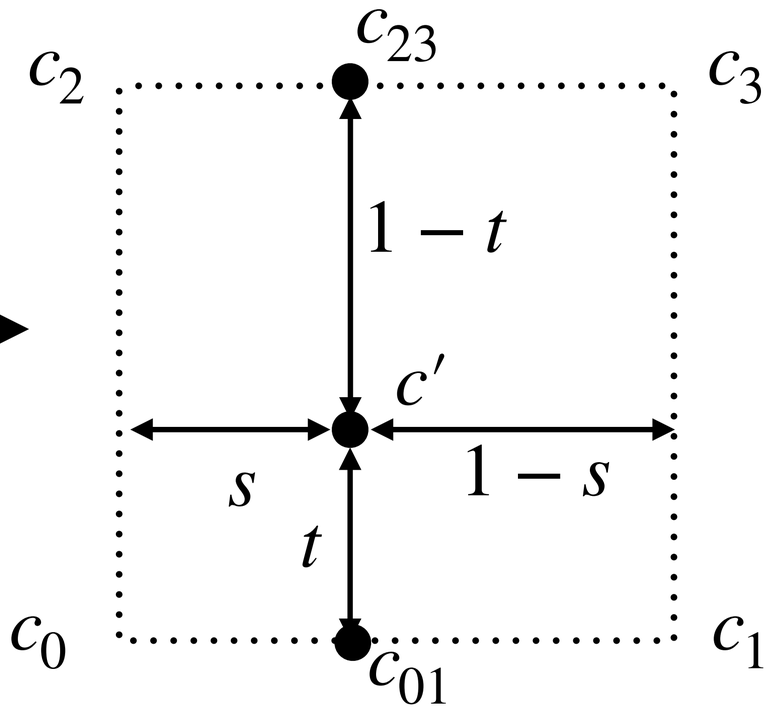
Linear Filtering



Nearest



Linear

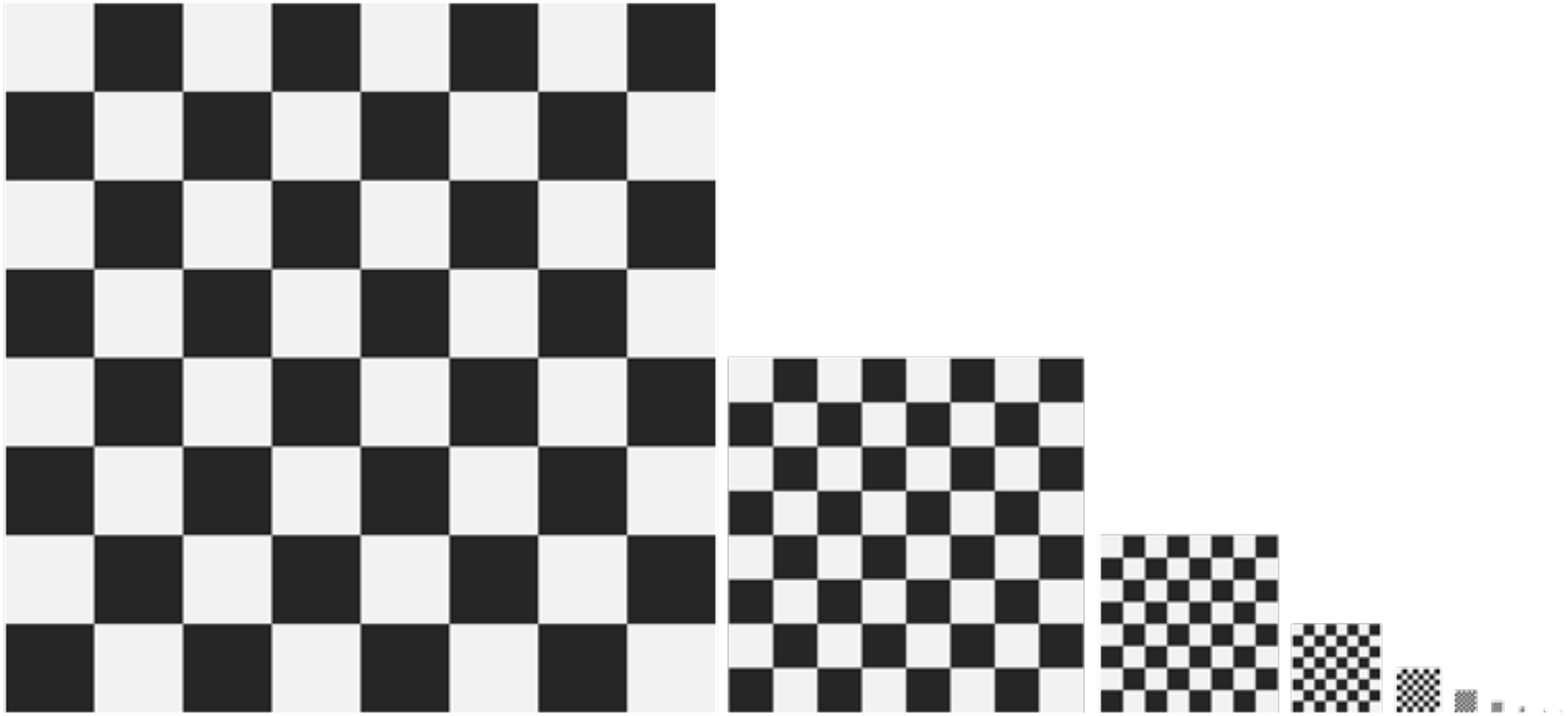


$$c_{01} = (1-s)\cdot c_0 + s\cdot c_1$$

$$c_{23} = (1-s)\cdot c_2 + s\cdot c_3$$

$$c' = (1-t)\cdot c_{01} + t\cdot c_{23}$$

Mipmap

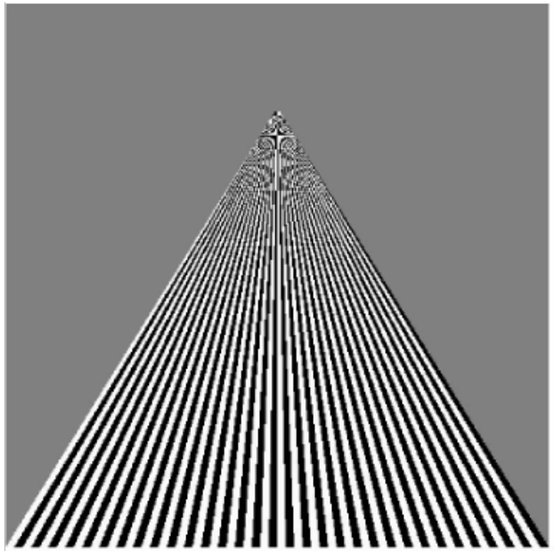


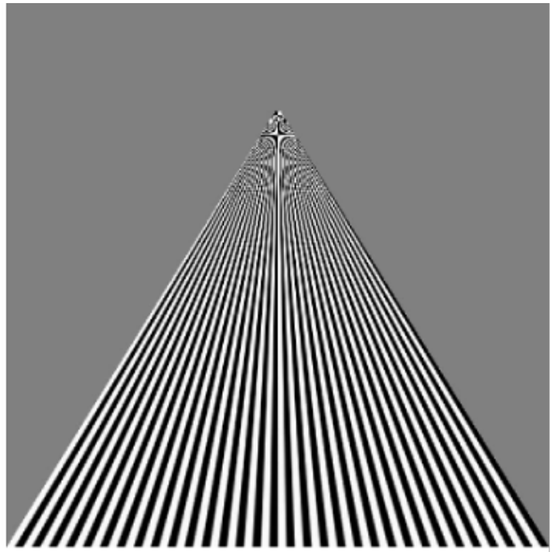
[source: arm developer]

```
gl.generateMipmap(gl.TEXTURE_2D);
gl.texParameteri(gl.TEXTURE_2D, gl.TEXTURE_MIN_FILTER,
                 gl.LINEAR_MIPMAP_LINEAR);
```

Results

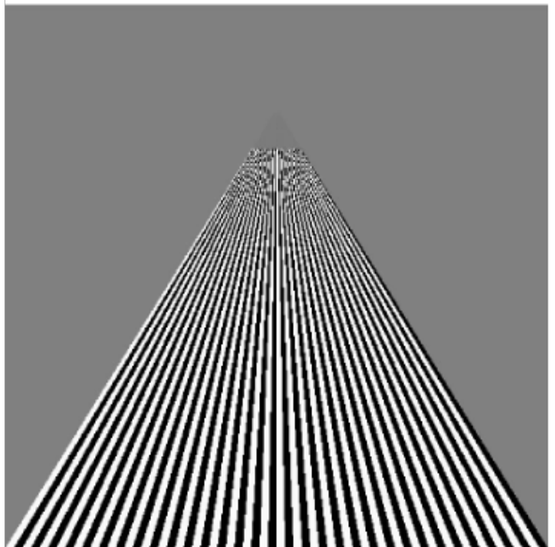
point sampling

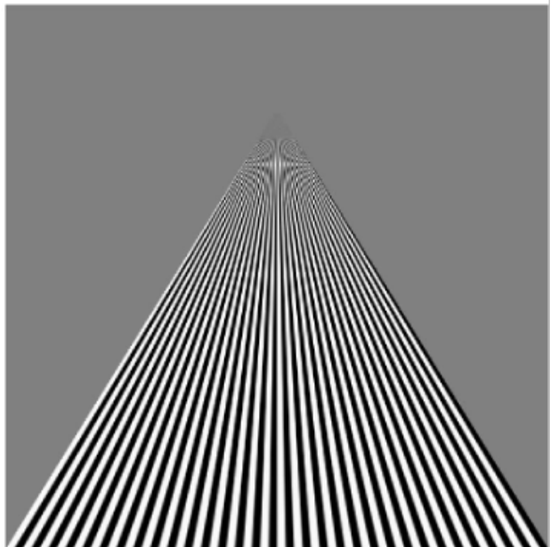




linear filtering

mipmapped point sampling





mipmapped linear filtering

[source: Prof. Edward Angel]